Streaming Data Analytics

Preview of the part of the exam about Streaming Data Science

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Exam content

- **Questions** on all lectures about Streaming Data Science **to test**
  - **The breadth** of your knowledge
  - **The depth** of your knowledge

(6 points)

- **Exercises**
  - Given a python **code snippet** that addresses a streaming binary classification problem or a time series forecasting problem **comment** what the code tries to achieve and **fill in missing parts** explaining **why** you did so
  - Given a particular **problem/task/situation**, explain **how** you would solve it (methods, metrics, evaluation used) and comment **why**

(9 points)

Questions to test the **breadth** of your **TSA** knowledge

- [SDS.TSA.1] What is the i.i.d. assumption? In which ways do SML and TSA relax parts of this assumption? In your answer, explicitly discuss the roles of temporal dependence and concept drift.
- [SDS.TSA.2] Which characteristics can a non-stationary time series have? For each type of non-stationarity you list, provide a corresponding example that illustrates it.
- [SDS.TSA.3] Why is the white noise a predictable time series? What is its forecast? What is its error?
- [SDS.TSA.4] Which are the methods to test for stationarity? List all the methods and explain one of them in detail.
- [SDS.TSA.5] Which are the typical time series components? Describe their properties and illustrate your explanation with an example.
- [SDS.TSA.6] Which are the methods to detrend a time series? ? List all the methods and explain one of them in detail.
- [SDS.TSA.7] Which are the methods to remove seasonality in a time series? List all the methods and explain one of them in detail.
- [SDS.TSA.8] Which are the benefits and limitations of [MAE / MAPE / MSE / RMSE]? Provide a practical example where its use is beneficial and one where it should be avoided.
- [SDS.TSA.9] What can a SARIMAX model account for in time series data? Describe its main capabilities and then explain the purpose of each of its components.
- [SDS.TSA.10] Describe how Prophet works and what each of its key components does. Highlight at least three advantages of using Prophet in time series forecasting.

Questions to test the **depth** of your **TSA** knowledge

- [SDS.TSA.11] What is the difference between an additive and a multiplicative model in time series decomposition? Discuss also when it is more appropriate to use one instead of the other.
- [SDS.TSA.12] Why is exponential smoothing named in this way? Discuss the formula of at least one of the three methods presented in the course.
- [SDS.TSA.13] What is the difference between simple, double, and triple exponential smoothing? Illustrate the explanation by highlighting the respective components in the triple exponential smoothing formula.
- [SDS.TSA.14] What is the difference between the meaning of moving average in time series decomposition and the MA component in ARMA models?
- [SDS.TSA.15] What is the definition of Autocorrelation? How does it differ from the definition of correlation?
- [SDS.TSA.16] What is the difference between the AR and the MA part of an ARMA model? What is their relation to Autocorrelation and Partial Autocorrelation? Illustrate your explanation using ACF and PACF plots.
- [SDS.TSA.17] How does the Box-Jenkins methodology guide the selection of the orders of an ARMA model? Describe the step-by-step procedure used to identify and refine the model order.
- [SDS.TSA.18] What are the exogenous variables in TSA? How do they contribute to the forecasting? What is their impact on the confidence interval? Illustrate your explanation with an example.

Questions to test the **breadth** of your SML knowledge

- [SDS.SML.1] What are the differences between batch-oriented Machine Learning and Streaming Machine Learning?
- [SDS.SML.2] What are the benefits and the challenges of Streaming Machine Learning?
- [SDS.SML.3] What is a drift? Which are the types of drift? Why is it so important to detect it? Illustrate the difference using the Bayes Theorem and with an example.
- [SDS.SML.4] How can you classify a concept drift with respect to the speed of change? Illustrate the difference with an example for each type.
- [SDS.SML.5] Which are the typical Streaming Machine Learning algorithms for classification? Illustrate one of them in detail.
- [SDS.SML.6] What are the key factors to consider when building an SML Ensemble Classification model?
- [SDS.SML.7] How does SML regression compare to time series forecasting w.r.t. types of input features, training, forecasting horizon, and adaptability?

Questions to test the **depth** of your **SML** knowledge

- [SDS.SML.8] Which are the drift detectors that monitor the error rate? Illustrate how one of them works.
- [SDS.SML.9] Which are the data drift detectors? Illustrate how one of them works.
- [SDS.SML.10] How does ADWIN detect a concept drift? Illustrate it with an example.
- [SDS.SML.11] How does the SML version of KNN work? Illustrate it with an example.
- [SDS.SML.12] How does the SML version of Naïve Bayes work? Illustrate it with an example.
- [SDS.SML.13] How does the Hoeffding Tree work? Illustrate it with an example.
- [SDS.SML.14] How does the Hoeffding Adaptive Tree work? Illustrate it with an example.
- [SDS.SML.15] What is Hoeffding Bound? How and why is it used in SML to build a decision tree?
- [SDS.SML.16] Why is the Poisson distribution used in SML ensemble methods? What’s the impact of the value of lambda? Illustrate how lambda is used in the SML methods.

Questions to test the **breadth & depth** of your **CL** knowledge

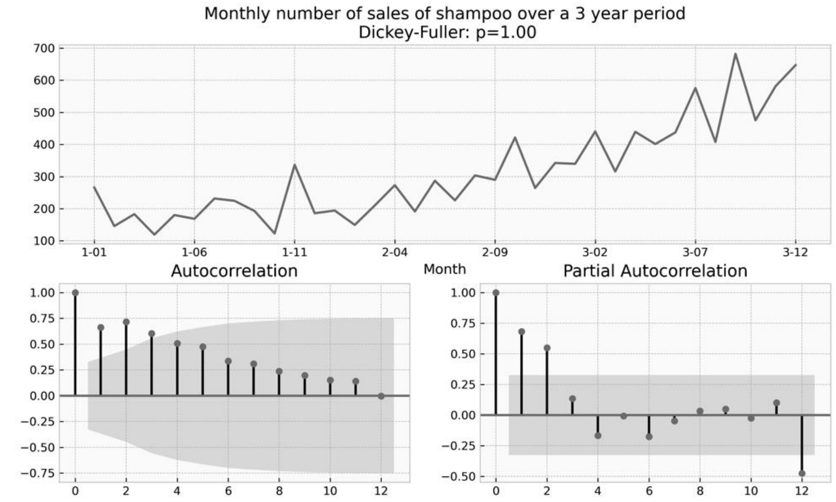
- [SDS.CL.1] What’s the stability-plasticity dilemma? Illustrate your explanation with a narrative example and discuss the different learning abilities that a CL solution should have.
- [SDS.CL.2] What are the main differences between Streaming Machine Learning and Continual Learning? Discuss them w.r.t to the setting, the goals, and the evaluation process.
- [SDS.CL.3] What are the main differences between Streaming Machine Learning and Continual Learning w.r.t. the concept drifts they manage? Discuss also which cases require avoiding forgetting and which do not.
- [SDS.CL.4] What are the three main scenarios in Continual Learning? Describe each of them briefly.
- [SDS.CL.5] How do CL replay strategies work? Explain one specific strategy in detail. What are the other two categories of CL strategies? Briefly explain them.
- [SDS.CL.6] How do CL regularization strategies work? Explain one specific strategy in detail. What are the other two categories of CL strategies? Briefly explain them.
- [SDS.CL.7] How do CL architectural strategies work? Explain one specific strategy in detail. What are the other two categories of CL strategies? Briefly explain them.
- [SDS.CL.8] Which are the primary evaluation metrics used in Continual Learning? Illustrate what they measure and why they are all useful to assess the properties of a method.

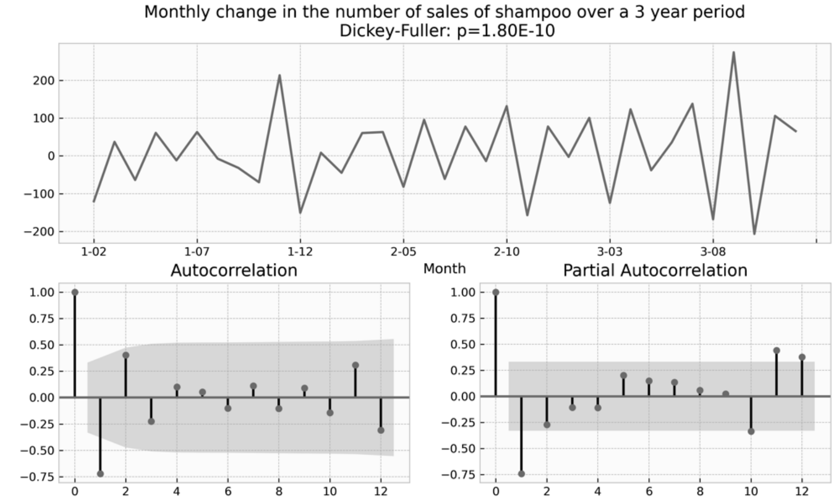
Exercises on TSA, part 1

Given a a code snippet that computes ACF and PACF plots illustrated:

- **Fill in the blank spaces**

```
1.  data_set = pd.read_csv(  'shampoo.csv',
2.                           infer_datetime_format=True,
3.                           parse_dates=["Month"],
4.                           index_col=["Month"])
    # ACF and PACF plots of monthly number
5.  ………………………………………(data_set['Sales'], lags= ………………, ax=acf_ax)
6.  ………………………………………(data_set['Sales'], lags= ………………, ax=pacf_ax)
    # ACF and PACF plots of monthly changes (month_i – month_i-1)
7.  y = ……………………………………………………………………
8.  ………………………………………(y, lags= ………………, ax=acf_ax)
9.  ………………………………………(y, lags= ………………, ax=pacf_ax)
10. plot.show()
```





Exercises on TSA, part 2

- **Comment each line of the code-snippet.**

1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
9.	
10.	

- **Using the provided ACF and PACF plots, and ADF test results for the shown time series, discuss a possible order for an ARIMA model.**
  - **Consider the ADF test p-value to guide the analysis between the two time series.**
  - **Analyze the ACF and PACF plots to determine the appropriate values for p and q, looking for cut-off and tail-off patterns to inform your choices.**

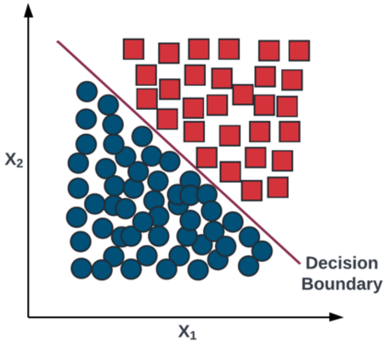
Exercises on SML

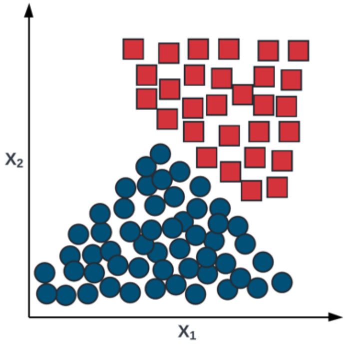
Given a data distribution plot, a code-snippet representing the algorithm training and testing phases, and the resulting error rate plot over time:

- Comment each line of the code-snippet;
- Discuss the ability of the proposed code to address the data distribution and the error rate illustrated in the two plots;
- Tell if a concept drift occurred, its type, and if it necessary to change the decision boundary;
- How would you modify the code to obtain … (a different situation w.r.t. the illustrated one).

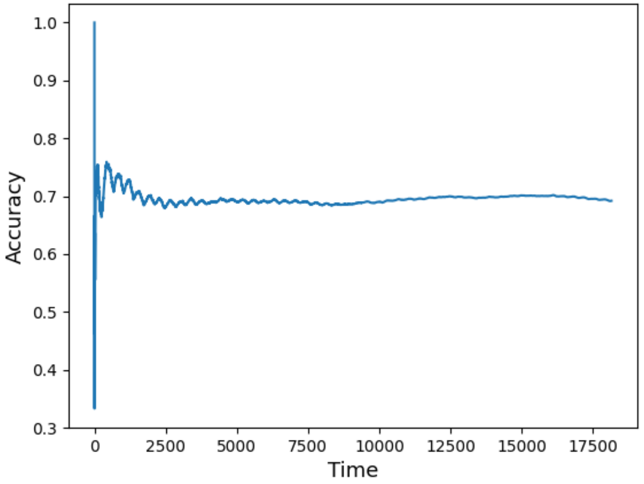
Solution 1/2

Given a data distribution plot, a code-snippet representing the algorithm training and testing phases, and the resulting error rate plot over time:





1. data = pd.read_csv("stream.csv")
2. features = data.columns[:-1]
3. label = data.columns[-1]
4. stream = iter_pandas(X=data[features], y=data[label])
5. model = GaussianNB()
6. metric = Accuracy()
7. progressive_val_score(dataset=stream,
   model=model,
   metric=metrics,
   print_every=1000)



Solution 2/2

- **Comment each line of the code-snippet**

1.	
2.	
3.	
4.	
5.	
6.	
7.	

- **Discuss the ability of the proposed code to address the data distribution and the error rate illustrated in the two plots**
- **Tell if a concept drift occurred, its type, and if it necessary to change the decision boundary;**
- **Choose one pipeline module to change to obtain more accurate and realistic results among model, change detector, metric, and type of evaluation. Discuss your answer**

Exercises on CL

Draw the table (or the performance plot) of a CL solution with the following learning abilities:

- Positive/Negative BWT
- Forgetting
- Positive/Negative FWT
- Plasticity/Absence of plasticity
- Stability/Absence of stability

See 22_CL_theory

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